Quiz: Generators and Iterators

Total Score: /2^3

Printed Name:

Quiz rules:

1. You MAY use any printed or handwritten notes.
2. You MAY NOT use a computer or any other electronic device.

Problem 1. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1

def take(g, n):
    for _ in range(n):
        yield next(g)
try:
    g = nats()
    print('take=', list(take(g, 3)))
    print('next(g)=', next(g))
except TypeError:
    print('TypeError')
except StopIteration:
    print('StopIteration')
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 15 / 16 = 0.94

Problem 2. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
xs = [1, 2, 3]
try:
    ys = map(lambda x: x * x, xs)
    print('list=', list(ys))
    print('len=', len(list(ys)))
except TypeError:
    print('TypeError')
except StopIteration:
    print('StopIteration')
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 3. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class CountdownIter:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i + 1

class Countdown:
    def __init__(self, n):
        self.n = n
    def __iter__(self):
        return CountdownIter(self.n)
try:
    c = Countdown(3)
    print('a=', list(c))
    print('b=', sum(c))
except TypeError:
    print('TypeError')
except StopIteration:
    print('StopIteration')
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 15 / 16 = 0.94

Problem 4. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
try:
    print('cycle=', list(itertools.islice(itertools.cycle([1, 2, 3]), 7)))
    print('sum=', sum(itertools.islice(itertools.cycle([4, 5]), 5)))
except TypeError:
    print('TypeError')
except StopIteration:
    print('StopIteration')
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 13 / 16 = 0.81

LLM Model Performance

